Supplementary Material:
Evaluating Tabular Representation Learning for Network Intrusion Detection

This supplementary material contains comprehensive tables with the full experimental results for our study across multiple benchmark datasets using various representation learning techniques.

Dataset Class Distribution

This section presents the class distribution statistics for the datasets used in our experiments.

Table 1: UNSW-NB15 Dataset Class Distribution

Class Name	**Count**	**Percentage**
Benign	2,295,222	96.02%
Exploits	31,551	1.32%
Fuzzers	22,310	0.93%
Generic	16,560	0.69%
Reconnaissance	12,779	0.53%
DoS	5,794	0.24%
Analysis	2,299	0.10%
Backdoor	2,169	0.09%
Shellcode	1,427	0.06%
Worms	164	0.01%
Total	**2,390,275**	**100.00%**

Table 2: CIDDS Dataset Class Distribution

Class Name	**Count**	**Percentage**
Normal	28,051,906	89.66%
Attacker	1,656,605	5.29%
Victim	1,579,422	5.05%
Total	**31,287,933**	**100.00%**

Table 3: NF-CSE-CIC-IDS2018-v2 Dataset Class Distribution

Class Name	**Count**	**Percentage**
Benign	16,635,567	88.05%
DDOS attack-HOIC	1,080,858	5.72%
DoS attacks-Hulk	432,648	2.29%
DDoS attacks-LOIC-HTTP	307,300	1.63%
Bot	143,097	0.76%
Infilteration	116,361	0.62%
SSH-Bruteforce	94,979	0.50%
DoS attacks-GoldenEye	27,723	0.15%
FTP-BruteForce	25,933	0.14%
DoS attacks-SlowHTTPTest	14,116	0.07%
DoS attacks-Slowloris	9,512	0.05%
Brute Force -Web	2,143	0.01%
DDOS attack-LOIC-UDP	2,112	0.01%
Brute Force -XSS	927	0.00%
SQL Injection	432	0.00%
Total	**18,893,708**	**100.00%**

Full Tables for All Models and Datasets

Table 5: Best Supervised and Unsupervised Model Results on NF-UNSW-NB15-v2

Model	RL-Model	Configuration	AUCROC	AUCPR	F1-Score
PCA-based Models					
PCA	SCARF	contamination=0.1, svd_solver=randomized n_components=15	0.591	0.151	0.222
PCA	TabICL	contamination=0.25, svd_solver=randomized n_components=5	0.976	0.481	0.488
PCA	CC	contamination=0.1, svd_solver=auto n_components=5	0.835	0.009	0.015
PCA	Autoencoder	contamination=0.1, svd_solver=auto n_components=5	0.948	0.615	0.540
One-Class SVM Models					
OCSVM	SCARF	nu=0.9, degree=2 gamma=scale	0.526	0.114	0.234
OCSVM	TabICL	nu=0.5, degree=2 gamma=scale	0.856	0.387	0.175
OCSVM	CC	nu=0.1, degree=2 gamma=auto	0.878	0.018	0.002
OCSVM	Autoencoder	nu=0.1, degree=3 gamma=scale	0.448	0.185	0.274
Isolation Forest Models					
IForest	SCARF	n_estimators=200, max_samples=100, max_features=0.5, contamination=0.5	0.612	0.156	0.248
IForest	TabICL	n_estimators=200, max_samples=100, max_features=0.7, contamination=0.1	0.974	0.470	0.482
IForest	CC	n_estimators=200, max_samples=50, max_features=0.7, contamination=0.5	0.854	0.008	0.021
IForest	Autoencoder	n_estimators=200, max_samples=50, max_features=0.5, contamination=0.1	0.926	0.545	0.425
K-Nearest Neighbors Models					
KNN	SCARF	n_neighbors=5, algorithm=auto, metric=euclidean	0.674	0.075	0.050
KNN	TabICL	n_neighbors=10, algorithm=auto, metric=manhattan	0.917	0.346	0.229
KNN	CC	n_neighbors=3, algorithm=auto, metric=minkowski	0.882	0.028	0.001
KNN	Autoencoder	n_neighbors=5, algorithm=auto, metric=euclidean	0.642	0.198	0.300
Support Vector Machine Models					
SVM	SCARF	gamma=scale, coef0=0.5, tol=0.1, C=1.0, kernel=linear	0.764	0.404	0.447
SVM	TabICL	gamma=scale, coef0=0.0, tol=0.0001, C=0.1, kernel=linear	0.991	0.970	0.045
SVM	CC	gamma=scale, coef0=0.0, tol=0.0001, C=0.1, kernel=rbf	0.896	0.094	0.005
SVM	Autoencoder	gamma=scale, coef0=0.0, tol=0.0001, C=10, kernel=rbf	0.957	0.892	0.801
Logistic Regression Models					
LR	SCARF	penalty=l2, C=0.1, class_weight=none, solver=lbfgs	0.873	0.304	0.072
LR	TabICL	penalty=l2, C=0.1, class_weight=none, solver=newton-cg	0.879	0.546	0.484
LR	CC	penalty=l2, C=10, class_weight=none, solver=liblinear	0.973	0.681	0.114
LR	Autoencoder	penalty=l2, C=0.1, class_weight=balanced, solver=sag	0.979	0.860	0.001
Random Forest Models					
RF	SCARF	n_estimators=200, max_depth=50, class_weight=balanced, ccp_alpha=0.01	0.806	0.157	0.077
RF	TabICL	n_estimators=200, max_depth=10, class_weight=balanced, ccp_alpha=0.1	0.987	0.844	0.700
RF	CC	n_estimators=10, max_depth=10, class_weight=None, ccp_alpha=0	0.899	0.427	0.004
RF	Autoencoder	n_estimators=100, max_depth=10, class_weight=balanced, ccp_alpha=0.01	0.987	0.934	0.801
XGBoost Models					
XGBoost	SCARF	lr=0.01, max_depth=3, n_estimators=100, reg_alpha=0.5, reg_lambda=1	0.850	0.317	0.077
XGBoost	TabICL	lr=0.1, max_depth=3, n_estimators=500, reg_alpha=0, reg_lambda=0.5	0.958	0.505	0.001
XGBoost	CC	lr=0.01, max_depth=5, n_estimators=1000, reg_alpha=0.5, reg_lambda=0.1	0.990	0.664	0.004
XGBoost	Autoencoder	lr=0.01, max_depth=7, n_estimators=100, reg_alpha=0, reg_lambda=0.1	0.977	0.836	0.796
End-to-End Deep Learning Models					
BERT	End-to-End (classification head)	lr=1e-5, batch_size=32, epochs=4	0.996	0.894	0.834
Autoencoder	End-to-End (classification head)	latent_dim=64	0.991	0.763	0.722

Table 4: Best Supervised and Unsupervised Model Results on CIDDS-001

Model	RL-Model	Configuration	AUCROC	AUCPR	F1-Score
PCA-based Models					
PCA	SCARF	contamination=0.5, svd_solver=auto n_components=10	0.534	0.166	0.398
PCA	TabICL	contamination=0.25, svd_solver=randomized n_components=5	0.642	0.252	0.237
PCA	CC	contamination=0.1, svd_solver=auto n_components=5	0.369	0.001	0.000
PCA	Autoencoder	contamination=0.5, svd_solver=randomized n_components=5	0.926	0.538	0.473
One-Class SVM Models					
OCSVM	SCARF	nu=0.9, degree=2 gamma=auto	0.625	0.222	0.315
OCSVM	TabICL	nu=0.5, degree=2 gamma=scale	0.276	0.115	0.174
OCSVM	CC	nu=0.1, degree=2 gamma=scale	0.413	0.001	0.003
OCSVM	Autoencoder	nu=0.1, degree=2 gamma=scale	0.376	0.168	0.281
Isolation Forest Models					
IForest	SCARF	n_estimators=200, max_samples=50, max_features=0.7, contamination=0.5	0.652	0.246	0.350
IForest	TabICL	n_estimators=200, max_samples=50, max_features=1.0, contamination=0.5	0.550	0.170	0.136
IForest	CC	n_estimators=200, max_samples=50, max_features=0.7, contamination=0.5	0.399	0.001	0.002
IForest	Autoencoder	n_estimators=200, max_samples=50, max_features=0.7, contamination=0.25	0.958	0.694	0.791
K-Nearest Neighbors Models					
KNN	SCARF	n_neighbors=10, algorithm=auto, metric=euclidean	0.677	0.251	0.198
KNN	TabICL	n_neighbors=3, algorithm=auto, metric=cosine	0.412	0.136	0.150
KNN	CC	n_neighbors=3, algorithm=auto, metric=manhattan	0.540	0.001	0.003
KNN	Autoencoder	n_neighbors=5, algorithm=brute, metric=manhattan	0.637	0.198	0.316
Support Vector Machine Models					
SVM	SCARF	gamma=scale, coef0=0.5, tol=0.1, C=1.0, kernel=linear	0.764	0.404	0.447
SVM	TabICL	gamma=scale, coef0=0.0, tol=0.0001, C=0.1, kernel=linear	0.991	0.970	0.045
SVM	CC	gamma=scale, coef0=0.0, tol=0.0001, C=0.1, kernel=rbf	0.896	0.094	0.005
SVM	Autoencoder	gamma=scale, coef0=0.0, tol=0.0001, C=0.1, kernel=linear	0.970	0.768	0.936
Logistic Regression Models					
LR	SCARF	penalty=l2, C=0.1, class_weight=balanced, solver=lbfgs	0.914	0.665	0.389
LR	TabICL	penalty=l2, C=0.1, class_weight=balanced, solver=liblinear	0.997	0.991	0.728
LR	CC	penalty=none, C=0.1, class_weight=none, solver=newton-cg	0.956	0.236	0.059
LR	Autoencoder	penalty=l2, C=10, class_weight=none, solver=lbfgs	0.980	0.943	0.739
Random Forest Models					
RF	SCARF	n_estimators=200, max_depth=50, class_weight=balanced, ccp_alpha=0.01	0.869	0.625	0.614
RF	TabICL	n_estimators=200, max_depth=10, class_weight=balanced, ccp_alpha=0.1	0.979	0.945	0.883
RF	CC	n_estimators=200, max_depth=50, class_weight=balanced, ccp_alpha=0	0.928	0.524	0.515
RF	Autoencoder	n_estimators=200, max_depth=10, class_weight=balanced, ccp_alpha=0	0.986	0.930	0.882
XGBoost Models					
XGBoost	SCARF	lr=0.01, max_depth=3, n_estimators=100, reg_alpha=0, reg_lambda=1	0.994	0.977	0.745
XGBoost	TabICL	lr=0.1, max_depth=5, n_estimators=100, reg_alpha=0, reg_lambda=0.1	0.940	0.719	0.171
XGBoost	CC	lr=0.01, max_depth=5, n_estimators=1000, reg_alpha=0.5, reg_lambda=0.1	0.972	0.449	0.429
XGBoost	Autoencoder	lr=0.01, max_depth=3, n_estimators=100, reg_alpha=0, reg_lambda=0.1	0.911	0.733	0.787
End-to-End Deep Learning Models					
BERT	End-to-End (classification head)	lr=1e-5, batch_size=32, epochs=4	0.986	0.794	0.864
Autoencoder	End-to-End (classification head)	latent_dim=16	0.963	0.733	0.608

Table 6: Best Supervised and Unsupervised Model Results on NF-CSE-CIC-IDS2018

Model	RL-Model	Configuration	AUCROC	AUCPR	F1-Score
PCA-based Models					
PCA	SCARF	contamination=0.5, svd_solver=randomized n_components=5	0.604	0.156	0.116
PCA	TabICL	contamination=0.25, svd_solver=randomized n_components=5	0.591	0.151	0.221
PCA	CC	contamination=0.1, svd_solver=auto n_components=5	0.835	0.009	0.015
PCA	Autoencoder	contamination=0.1, svd_solver=randomized n_components=5	0.600	0.153	0.145
One-Class SVM Models					
OCSVM	SCARF	nu=0.9, degree=2 gamma=scale	0.527	0.114	0.234
OCSVM	TabICL	nu=0.9, degree=2 gamma=scale	0.793	0.342	0.084
OCSVM	CC	nu=0.1, degree=2 gamma=auto	0.492	0.143	0.201
OCSVM	Autoencoder	nu=0.1, degree=2 gamma=auto	0.518	0.122	0.213
Isolation Forest Models					
IForest	SCARF	n_estimators=200, max_samples=100, max_features=0.1, contamination=1.0	0.626	0.154	0.244
IForest	TabICL	n_estimators=200, max_samples=100, max_features=0.5, contamination=0.5	0.612	0.156	0.248
IForest	CC	n_estimators=200, max_samples=50, max_features=0.7, contamination=0.5	0.562	0.152	0.231
IForest	Autoencoder	n_estimators=200, max_samples=50, max_features=1.0, contamination=0.1	0.656	0.168	0.235
K-Nearest Neighbors Models					
KNN	SCARF	n_neighbors=10, algorithm=auto, metric=euclidean	0.704	0.206	0.120
KNN	TabICL	n_neighbors=10, algorithm=auto, metric=manhattan	0.491	0.106	0.099
KNN	CC	n_neighbors=10, algorithm=auto, metric=manhattan	0.344	0.107	0.294
KNN	Autoencoder	n_neighbors=3, algorithm=brute, metric=manhattan	0.234	0.079	0.238
Support Vector Machine Models					
SVM	SCARF	gamma=scale, coef0=0.5, tol=0.0001, C=1.0, kernel=poly	0.500	0.118	0.212
SVM	TabICL	gamma=scale, coef0=0.5, tol=0.0001, C=0.1, kernel=sigmoid	0.500	0.119	0.213
SVM	CC	gamma=scale, coef0=1.0, tol=0.0001, C=10, kernel=sigmoid	0.741	0.094	0.946
SVM	Autoencoder	gamma=scale, coef0=0.0, tol=0.0001, C=10, kernel=rbf	0.957	0.892	0.801
Logistic Regression Models					
LR	SCARF	penalty=l2, C=0.1, class_weight=none, solver=saga	0.642	0.247	0.209
LR	TabICL	penalty=l2, C=0.1, class_weight=balanced, solver=sag	0.881	0.405	0.579
LR	CC	penalty=l2, C=10, class_weight=none, solver=sag	0.946	0.913	0.215
LR	Autoencoder	penalty=l2, C=0.1, class_weight=none, solver=liblinear	0.887	0.815	0.204
Random Forest Models					
RF	SCARF	n_estimators=10, max_depth=50, class_weight=balanced, ccp_alpha=0.01	0.868	0.627	0.479
RF	TabICL	n_estimators=200, max_depth=50, class_weight=balanced, ccp_alpha=0.01	0.971	0.922	0.850
RF	CC	n_estimators=10, max_depth=10, class_weight=None, ccp_alpha=0	0.729	0.497	0.216
RF	Autoencoder	n_estimators=100, max_depth=10, class_weight=balanced, ccp_alpha=0.1	0.930	0.826	0.388
XGBoost Models					
XGBoost	SCARF	lr=0.01, max_depth=3, n_estimators=1000, reg_alpha=0, reg_lambda=1	0.763	0.430	0.214
XGBoost	TabICL	lr=0.01, max_depth=3, n_estimators=100, reg_alpha=0, reg_lambda=0.1	0.886	0.443	0.587
XGBoost	CC	lr=0.01, max_depth=5, n_estimators=1000, reg_alpha=0.5, reg_lambda=0.1	0.891	0.611	0.640
XGBoost	Autoencoder	lr=0.01, max_depth=7, n_estimators=100, reg_alpha=0.1, reg_lambda=0.1	0.800	0.550	0.145
End-to-End Deep Learning Models					
BERT	End-to-End (classification head)	lr=2e-5, batch_size=16, epochs=3	0.9997	0.938	0.883
Autoencoder	End-to-End (classification head)	latent_dim=8	0.975	0.903	0.876

Full Performance Ranking Tables

Table 7: F1-Score Rankings of Supervised Models Across Datasets

Method	CIDDS Rank	UNSW Rank	CSE-CIC Rank	Avg. Rank
BERT	4	1	2	2.33
SVM + Autoencoder	1	2	4	2.33
RF + TabICL	2	5	3	3.33
RF + Autoencoder	3	3	9	5.00
LR + TabICL	8	6	7	7.00
RF + SCARF	9	9	8	8.67
XGBoost + Autoencoder	5	4	17	8.67
XGBoost + SCARF	6	10	12	9.33
SVM + CC	17	13	1	10.33
SVM + SCARF	11	7	14	10.67
XGBoost + CC	12	15	5	10.67
LR + CC	15	8	11	11.33
RF + CC	10	14	10	11.33
XGBoost + TabICL	14	17	6	12.33
LR + Autoencoder	7	16	16	13.00
LR + SCARF	13	11	15	13.00
SVM + TabICL	16	12	13	13.67

Table 8: AUC-ROC Rankings of Supervised Models Across Datasets

Method	CIDDS Rank	UNSW Rank	CSE-CIC Rank	Avg. Rank
BERT	5	1	1	2.33
RF + Autoencoder	4	4	5	4.33
RF + TabICL	7	5	2	4.67
XGBoost + CC	8	3	6	5.67
LR + Autoencoder	6	6	7	6.33
SVM + TabICL	3	2	16	7.00
LR + CC	10	8	4	7.33
SVM + Autoencoder	9	10	3	7.33
LR + TabICL	1	13	9	7.67
XGBoost + TabICL	11	9	8	9.33
XGBoost + SCARF	2	15	12	9.67
XGBoost + Autoencoder	14	7	11	10.67
RF + CC	12	11	14	12.33
SVM + CC	15	12	13	13.33
LR + SCARF	13	14	15	14.00
RF + SCARF	16	16	10	14.00
SVM + SCARF	17	17	17	17.00

Table 9: AUC-ROC Rankings of Unsupervised Models Across Datasets

Method	**CIDDS Rank**	**UNSW Rank**	**CSE Rank**	**Avg. Rank**
IForest + Autoencoder	1	4	4	3.00
PCA + Autoencoder	2	3	8	4.33
PCA + TabICL	5	1	9	5.00
IForest + TabICL	8	2	6	5.33
KNN + SCARF	3	11	3	5.67
IForest + SCARF	4	13	5	7.33
OCSVM + TabICL	16	8	2	8.67
PCA + CC	15	10	1	8.67
KNN + CC	9	6	15	10.00
KNN + TabICL	12	5	14	10.33
OCSVM + CC	11	7	13	10.33
PCA + SCARF	10	14	7	10.33
IForest + CC	13	9	10	10.67
OCSVM + SCARF	7	15	11	11.00
KNN + Autoencoder	6	12	16	11.33
OCSVM + Autoencoder	14	16	12	14.00

Cross-Dataset Transfer Learning Results

This section presents the results from the cross-dataset transfer learning experiments, where models trained on one dataset are evaluated on another selecting the best model configuration (according to validation AUC-PR) for each transfer path.

Table 10: Cross-Dataset Transfer Learning Results (Supervised)

Method	Source	Target	Model	AUCROC	AUCPR
CC	CIDDS	CSE-CIC	SVM	0.9185	0.6108
CC	CIDDS	UNSW-NB15	SVM	0.7995	0.5701
CC	CSE-CIC	CIDDS	SVM	0.6110	0.3548
CC	CSE-CIC	UNSW-NB15	SVM	0.6107	0.3811
CC	UNSW-NB15	CIDDS	LR	0.7514	0.4983
CC	UNSW-NB15	CSE-CIC	SVM	0.8733	0.6436
SCARF	CIDDS	CSE-CIC	LR	0.9029	0.7038
SCARF	CIDDS	UNSW-NB15	SVM	0.8567	0.5733
SCARF	CSE-CIC	CIDDS	SVM	0.9514	0.7992
SCARF	CSE-CIC	UNSW-NB15	LR	0.7267	0.3923
SCARF	UNSW-NB15	CIDDS	SVM	0.9315	0.7741
SCARF	UNSW-NB15	CSE-CIC	LR	0.9238	0.7856
TabICL	CIDDS	CSE-CIC	SVM	0.8642	0.6803
TabICL	CIDDS	UNSW-NB15	LR	0.5405	0.1469
TabICL	CSE-CIC	CIDDS	SVM	0.9637	0.9249
TabICL	CSE-CIC	UNSW-NB15	LR	0.8741	0.5547
TabICL	UNSW-NB15	CIDDS	SVM	0.9471	0.8983
TabICL	UNSW-NB15	CSE-CIC	LR	0.8978	0.7318

Table 11: Cross-Dataset Transfer Learning Results (Unsupervised)

Method	Source	Target	Model	AUCROC	AUCPR
CC	CIDDS	CSE-CIC	LOF	0.8249	0.5863
CC	CIDDS	UNSW-NB15	IForest	0.8320	0.4473
CC	CSE-CIC	CIDDS	IForest	0.8850	0.5187
CC	CSE-CIC	UNSW-NB15	IForest	0.8681	0.3565
CC	UNSW-NB15	CIDDS	IForest	0.9164	0.7359
CC	UNSW-NB15	CSE-CIC	PCA	0.7489	0.3222
SCARF	CIDDS	CSE-CIC	IForest	0.8056	0.5837
SCARF	CIDDS	UNSW-NB15	IForest	0.9658	0.5167
SCARF	CSE-CIC	CIDDS	IForest	0.9284	0.6728
SCARF	CSE-CIC	UNSW-NB15	LOF	0.7099	0.4009
SCARF	UNSW-NB15	CIDDS	IForest	0.8402	0.4624
SCARF	UNSW-NB15	CSE-CIC	IForest	0.7147	0.2814
TabICL	CIDDS	CSE-CIC	IForest	0.7854	0.4727
TabICL	CIDDS	UNSW-NB15	IForest	0.9693	0.5035
TabICL	CSE-CIC	CIDDS	IForest	0.8349	0.5083
TabICL	CSE-CIC	UNSW-NB15	IForest	0.9522	0.5139
TabICL	UNSW-NB15	CIDDS	IForest	0.7703	0.3786
TabICL	UNSW-NB15	CSE-CIC	IForest	0.7215	0.3124

Multiclass Classification Analysis

This section presents per-class precision and recall visualizations for multiclass classification across the three representation learning methods (SCARF, TabICL, and CC).

CIDDS-001 Dataset: Multiclass Performance

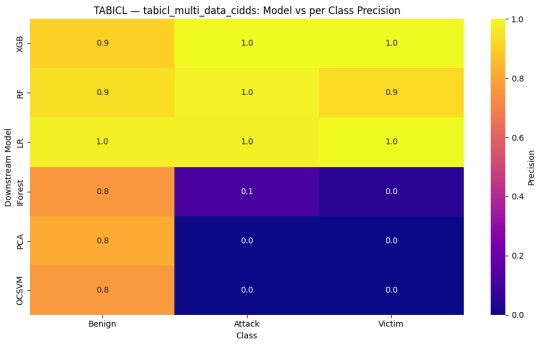

Figure 1: TabICL Precision per class on CIDDS-001

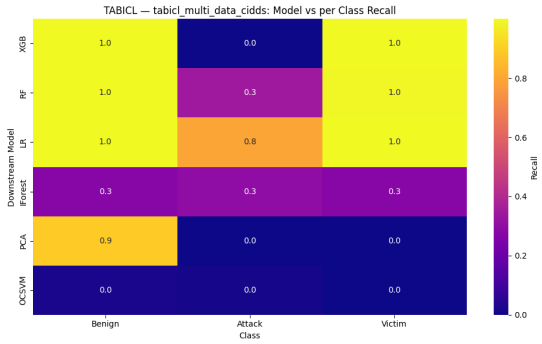

Figure 2: TabICL Recall per class on CIDDS-001

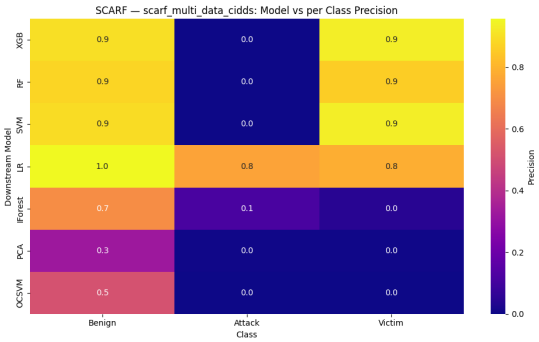

Figure 3: SCARF Precision per class on CIDDS-001

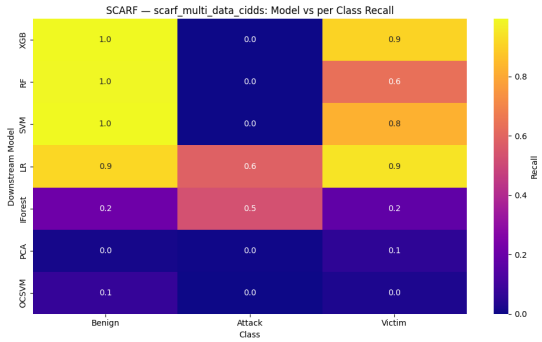

Figure 4: SCARF Recall per class on CIDDS-001

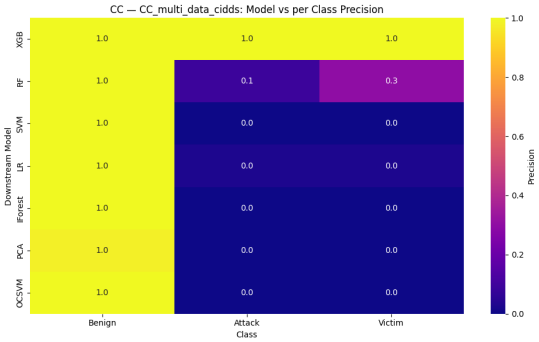

Figure 5: CC Precision per class on CIDDS-001

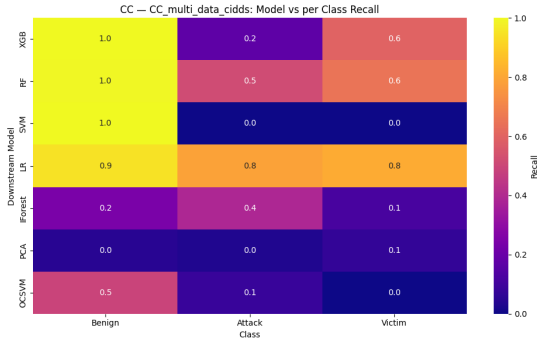

Figure 6: CC Recall per class on CIDDS-001

UNSW-NB15 Dataset: Multiclass Performance

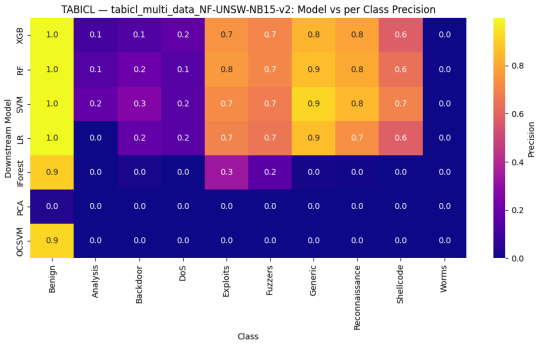

Figure 7: TabICL Precision per class on UNSW-NB15

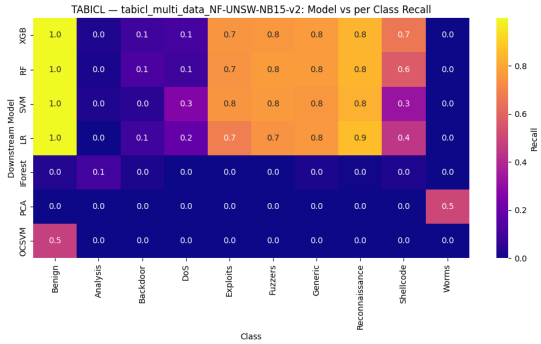

Figure 8: TabICL Recall per class on UNSW-NB15

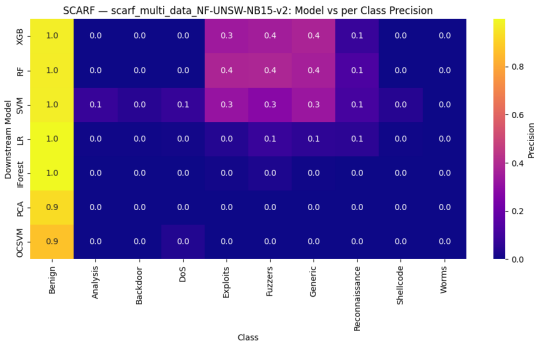

Figure 9: SCARF Precision per class on UNSW-NB15

Figure 10: SCARF Recall per class on UNSW-NB15

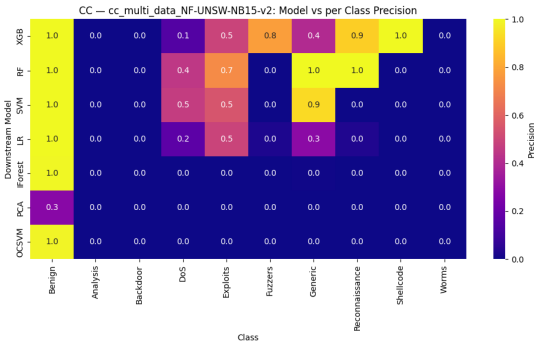

Figure 11: CC Precision per class on UNSW-NB15

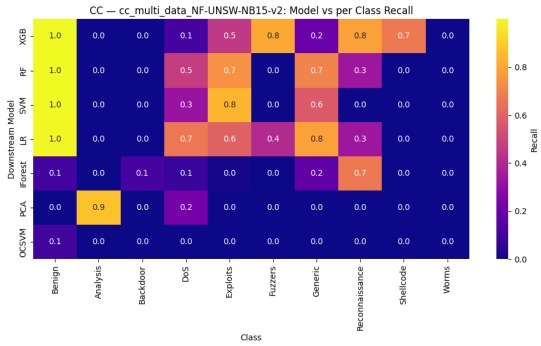

Figure 12: CC Recall per class on UNSW-NB15

CSE-CIC-IDS2018 Dataset: Multiclass Performance

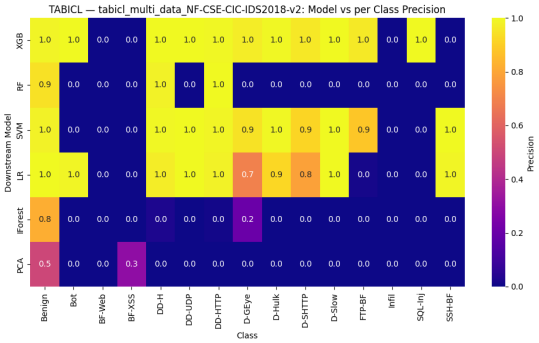

Figure 13: TabICL Precision per class on CSE-CIC-IDS2018

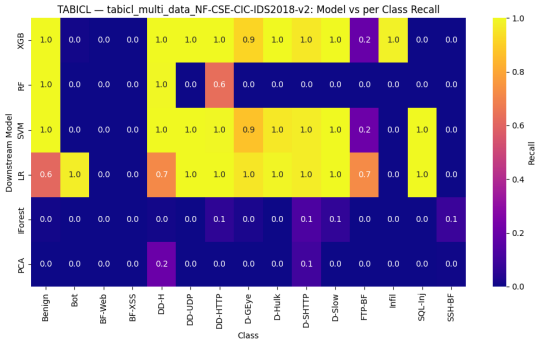

Figure 14: TabICL Recall per class on CSE-CIC-IDS2018

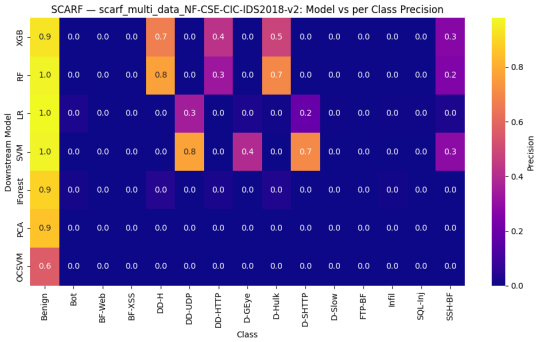

Figure 15: SCARF Precision per class on CSE-CIC-IDS2018

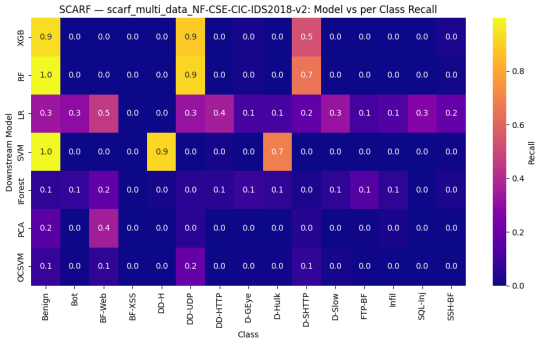

Figure 16: SCARF Recall per class on CSE-CIC-IDS2018

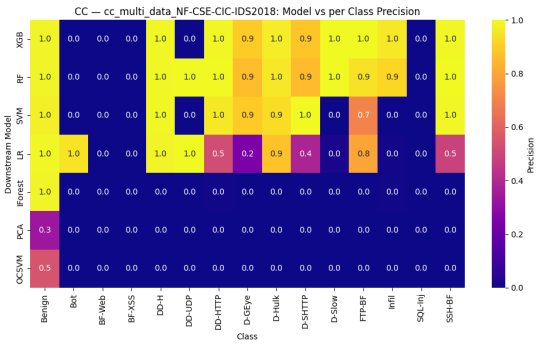

Figure 17: CC Precision per class on CSE-CIC-IDS2018

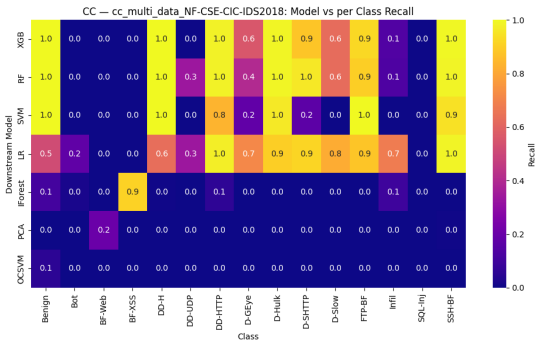

Figure 18: CC Recall per class on CSE-CIC-IDS2018

F1 Score and Macro F1 Analysis

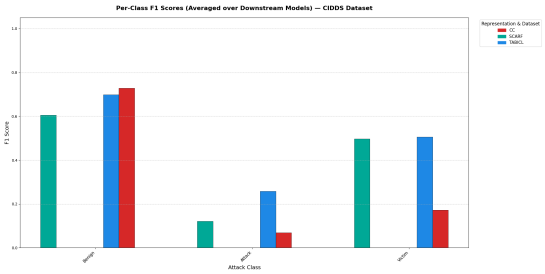

Figure 19: Per Class F1 score on CIDDS-001

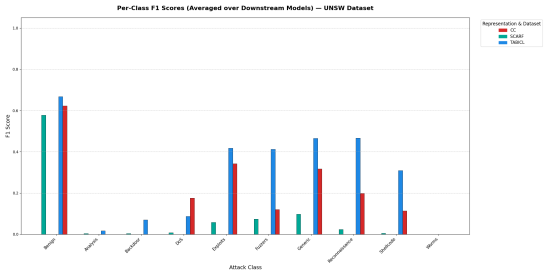

Figure 20: Per Class F1 score on UNSW-NB15

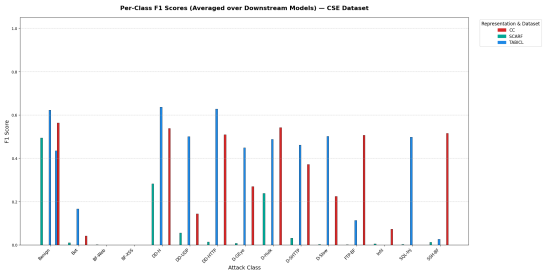

Figure 21: Per Class F1 score on CSE-CIC-IDS2018

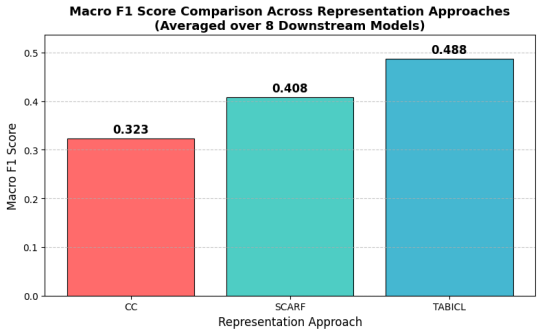

Figure 22: Average Macro F1 on CIDDS-001

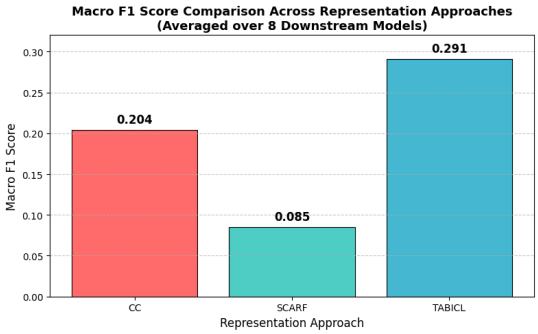

Figure 23: Average Macro F1 on UNSW-NB15

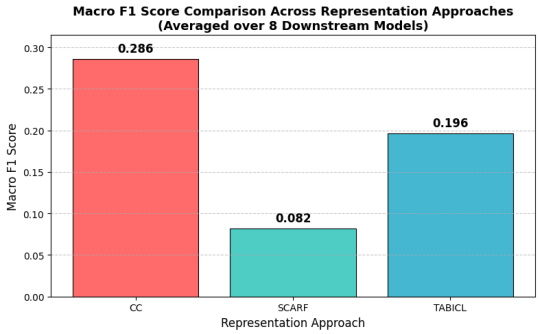

Figure 24: Average Macro F1 on CSE-CIC-IDS2018